Phishing Website Detection

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***Abstract*—This project showcases a method to classify websites into phishing vs. legitimate. Typical stages such as data cleaning, transformation, feature engineering are performed. Several models are created and tested. Using elastic-net, an accuracy score of approximately 99.74% is reached.**

Keywords—logistic regression, classification, phishing, elastic-net.

I. Introduction

Phishing is a common cyber attack technique that aims to steal information from unsuspecting users. Fake websites that resemble their real counterparts are created in hopes that some users will enter their personal and/or important financial information such as credit card numbers. This information is then sold and traded.

Creating a fake website is not a difficult task and may be done in a few steps. As such, a large number of these websites are created and distributed. While there are clear indicators, such as the domain name, it may escape the attention of the user. In this project, we aimed to create a classification algorithm that can act as a preliminary screening tool before the user attempts to load a website. This tool may warn users if the algorithm classifies the website as phishing.

II. Data Source

We used the PhiUSIIL dataset provided by the UC Irvine Machine Learning Repository. It includes about 235 thousand rows and 55 features. The target variable is a binary indicator of the legitimacy of the webpage. A label of 1 means a legitimate website.

The feature set can be divided into three categories: URL based features, HTML based features, and derived features. Descriptions of some of the features we deemed important are provided next.

A. URL Based Features

URL based features are extracted directly from the URLs of the websites.

Domain: The name of the domain address (e.g., metu.edu.tr).

TLD: Top level domain of the URL (e.g., .com, .edu), which is the last part of the domain name.

URL, TLD and Domain Length: Lengths of the URLs, TLDs, and the domain names.

Ratios of Characters in URL: Ratios of some special types of characters such as digits, letters, and special characters contained in the URL. Phishing URLs may contain more special characters to obscure their intentions.

Numbers of Characters in URL: Number of special characters, such as ampersand, equal sign, question mark, and other characters contained in the URL.

Number of Subdomains: The number of subdomains contained in the domain name.

Is Domain IP: Whether or not the website has a domain name or it contains an IP address instead. Phishing websites may not have registered a domain name.

Is HTTPS: Whether or not the website uses the secure HTTPS protocol instead of the less secure HTTP protocol.

B. HTML Based Features

When we connect to a website, we download an HTML file, which our browsers parse and display to us. These features are extracted from the HTML file.

Lines of CSS and JS code: The number of CSS and JavaScript code contained in the HTML output of the webpage.

Largest Line Length: Length of the longest line of code.

Number of Images: The number of images contained in the HTML output of the webpage.

Has Obfuscation: Whether or not the webpage uses obfuscation techniques to make its code less readable to humans.

Has Social Media Links: Whether or not the webpage includes links to its or others social media addresses.

Has Password Field: Whether or not the webpage includes a field to collect passwords from users.

Has Copyright/Description: Whether or not the webpage includes copyright and/or description information in its HTML output.

Has Favicon: Whether or not the webpage contains a webpage icon.

Has Title: Whether or not the webpage has a title element in its HTML output.

C. Derived Features

These features are derived from the previously discussed features. These features are provided by the creators of the dataset.

URL Similarity Index: Uses Levenstein's distance to compute the similarity between the URLs and a list of known legitimate URLs. An external, truthful source is used to achieve this purpose. Due to the definition of this feature, the legitimate URLs have perfect scores[6]

URL/Domain and Title Match Score: A measure of the difference between the URL/domain of a website and its title.

URL Character Probability: A measure of the randomness present in the URL.

III. Data Cleaning & Preprocessing

A. Dataset Construction Artifacts

The original dataset contained features derived from the URL such as `NoOfQMarkInURL`. As will be seen in the EDA step, some of these URL derived features are perfect predictors of the label, as in no legitimate URL contains special characters such as `?` or `=`, while phishing URLs often do. Upon further research, we have discovered that these are query characters and any legitimate URL can and do contain them [1]. A standard URL is composed of the following elements: a protocol, a domain name, a path with an optional query component. We have shown that the dataset does not include the path and the query component for legitimate URLs, which are the elements of a URL that contain special query characters. This is an issue of sampling by the original creators of the dataset and does not represent the reality. So, we have opted to drop the URL derived features completely and create similar features based on the domain name in the feature engineering section. Henceforth, the following features are dropped from the dataset:

- Features based on the length and the characters are derived from the path component. Thus, absence of the path in legitimate URLs biases these features, because the legitimate URLs do not contain the path component: `NoOfLettersInURL`, `LetterRatioInURL`, `NoOfDegitsInURL`, `DegitRatioInURL`, `URLLength`.

- Special characters are present in the path component of a URL, so the legitimate URLs do not contain them: `NoOfEqualsInURL`, `NoOfQMarkInURL`, `NoOfAmpersandInURL`, `NoOfOtherSpecialCharsInURL`, `SpacialCharRatioInURL`.

Next, we look at `TLD`. The TLD is the last part of the domain name (e.g., .com, .edu). When we counted the unique values of `TLD`, we encountered some issues. Consider the following URL:

```
https://www.365882.cc:8443/#/?sharename=3
65882.cc
```

The dataset lists its TLD as `cc:8443`. The number `8443`, represents the port number used to connect to the server, and it is not part of the TLD. Instead of using the original TLDs, we used the URL feature to extract the TLDs and replaced the original feature.

B. Categorical Feature Encoding

Most of the categorical features in the dataset are binary, which are already encoded using 0 and 1. As such, they do not require encoding. The remaining categorical features are strings. The `Title`, `Domain`, and `URL` features all contain unique values, so we do not encode them. We applied target encoding to `TLD`, since the feature has high cardinality and we wanted to capture its relationship with the target variable. We used the training set to compute the encoding rule and applied it on both the training and the test sets to avoid data leakage.

C. Handling Missing Data

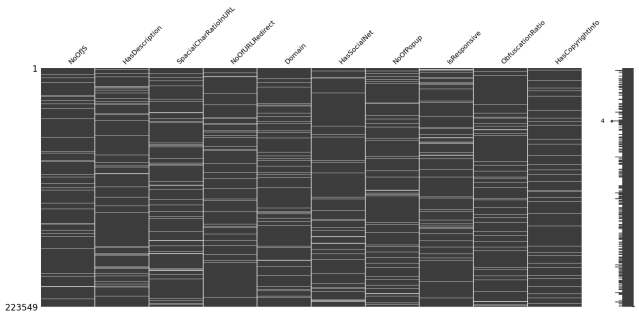

Fig 1. Visualization of missing data for a random subset of features.

Each feature in the dataset included approximately 5% missing data. We have provided the missing data visualization plot in Figure 1. All other features show similar missingness behaviour. Since the missingness does not seem to depend on other features in a row, we have concluded that it is MCAR, which is an ignorable class of missingness. We were not able to run Little's MCAR test due to the size of the dataset. Hence, we assume that the dataset exhibits MCAR.

We used MICE to impute missing values, because it is valid for MCAR and it takes into account the relationship between features [2]. However, we had some issues. MICE works with numerical values and does not respect the range of each future. For example, some of the binary variables had non-binary values after MICE. We applied some post imputation rules to constraint the values to desired ranges. We fitted the MICE model on the train set and used the results on the test set to avoid data leakage.

IV. Exploratory Data Analysis and Confirmatory Data Analysis

The dataset comprises features categorized into three main groups: URL-based structural attributes, HTML content indicators, and derived statistical metrics. These variables are designed to capture specific patterns associated with phishing attacks such as suspicious domain characteristics The target variable is a binary label that classifies each observation as either a legitimate (1) or a phishing (0) URL. Following the data cleaning process, a preliminary inspection revealed no significant anomalies in the dataset. Subsequently, variables were categorized into numeric, categorical, binary, and count types to facilitate appropriate visualization techniques for further analysis.

A. Count-Based Features

The analysis reveals distinct patterns across four key dimensions: URL structure, page complexity, connectivity, and obfuscation.

a. URL Structure and Composition

Significant disparities were observed in URL composition. Phishing URLs tend to be substantially longer (mean: approximately 45.5) compared to legitimate ones

(mean: approximately 26.2). Furthermore, specific characters such as equal signs (`NoOfEquals`) are frequently present in phishing URLs.

b. Page Complexity and Content

Legitimate websites exhibit a much higher level of complexity. Features such as `LineOfCode`, `NoOfImage`, `NoOfCSS`, and `NoOfJS` show significantly higher means and standard deviations in the legitimate class. This indicates that valid websites are content-rich and structurally developed.

c. Connectivity and References

Legitimate sites demonstrate a high volume of both external references (`NoOfExternalRef`) and self-references (`NoOfSelfRef`), indicative of a well-connected and navigable site structure. Conversely, phishing pages are relatively isolated, with fewer internal and external links.

d. Anomalies and Obfuscation

The `LargestLineLength` feature is notably higher in phishing instances. This is a strong indicator of code obfuscation, where attackers hide malicious scripts within extremely long lines of code to evade detection. Conversely, behavioral features like `NoSelfRedirect` and `NoOfURLRedirect` showed low variance across both classes, suggesting they may have limited predictive value for this specific dataset.

B. Binary-Based Features

Table I. Binary Features Means by Labels

Feature Name	Phishing (0)	Legitimate (1)
HasSocialNet	0.5%	79.4%
HasCopyrightInfo	5.7%	80.8%
HasDescription	4.3%	73.5%
HasSubmitButton	8.5%	66.2%

The Chi-Squared test identified `HasSocialNet`, `HasCopyrightInfo`, `HasDescription`, and `HasSubmitButton` as the most significant binary indicators. The analysis reveals a stark contrast in their distribution; for instance, `HasSocialNet` is present in 79.4% of legitimate websites but appears in only 0.5% of phishing URLs, suggesting that malicious sites typically lack standard functional elements. The complete distribution for the top features is detailed in Table I.

Table II. Binary Features Confusion Matrix Results

Feature	Precision	Specificity (TNR)	Sensitivity (TPR)	Youden's J
HasSocialNet	0.995	0.995	0.794	0.789
HasCopyrightInfo	0.950	0.943	0.808	0.751
HasDescription	0.959	0.957	0.735	0.693
HasSubmitButton	0.913	0.915	0.662	0.577

To further validate the discriminatory power of the selected binary features, a confusion matrix analysis was conducted. Each binary feature was evaluated as an independent classifier to assess its discriminatory power. As detailed in Table II, `HasSocialNet` emerged as the strongest predictor, achieving a near-perfect Precision (0.995) and the highest Youden's J score (0.789). Furthermore, the consistently high Specificity (>0.90) across all top features confirms that phishing websites rarely mimic these standard functional elements.

C. Numeric-Based Features

Univariate Analysis

Histograms and boxplots reveal that nearly all continuous variables exhibit non-normal, right-skewed distributions with widespread outliers. This pattern is characteristic of phishing datasets, where malicious URLs often represent statistical anomalies compared to legitimate URLs. Consequently, these extreme values and skewness will be addressed in the feature engineering phase to ensure model stability.

Bivariate Analysis: Correlation Matrix & Multicollinearity Check

The initial examination of the correlation matrix reveals significant sparsity, characterized by low correlations across the majority of features (indicated by the prevalence of neutral tones). Strong correlations are scarce and clustered within specific subsets. This suggests that the features largely provide independent information, which is advantageous for model performance by reducing redundancy.

To enhance interpretability and isolate the most significant relationships, a threshold-based filtering approach was applied. The resulting focused correlation matrix allows for a detailed inspection of potential multicollinearity.

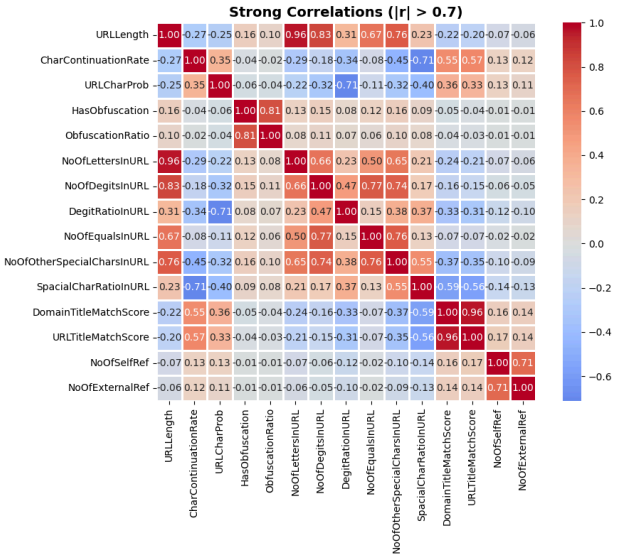

Fig 2. Correlation Matrix With 0.7 Threshold

The threshold-based correlation matrix as shown in Figure 2 reveals distinct clusters of multicollinearity. Notably, near-perfect positive correlations (approximately 0.96) were observed between structural pairs such as `URLLength`, and `NoOfLettersInURL`. This indicates significant redundancy, suggesting that these variable pairs convey nearly identical information. To maintain model stability, this multicollinearity will be systematically resolved in the feature engineering phase.

Linear Relationships with the Target Variable

Following the inter-feature analysis, the focus shifts to quantifying the direct impact of individual features on the target variable. To provide a clear overview, the analysis isolates the features with the strongest linear associations with the class label.

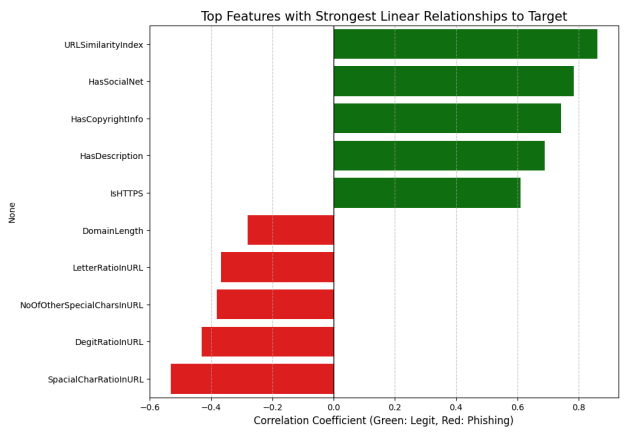

Fig 3. Features with Strong Linear Relationships to Target

As illustrated in the bar plot in Figure 3, green bars denote a positive correlation, indicating that an increase in these values is associated with legitimate websites. Conversely, red bars represent a negative correlation, serving as indicators for phishing attempts.The analysis highlights a distinct dichotomy: features such as `URLSimilarityIndex` and `HasSocialNet` demonstrate strong positive correlations with legitimate sites. Conversely, metrics reflecting structural irregularity, specifically high ratios of special characters and digits exhibit significant negative correlations, identifying them as primary indicators of phishing activity.

Investigation of `URLSimilarityIndex`

Although `URLSimilarityIndex` exhibited the highest correlation with the target variable, preliminary inspection suggested an unusually perfect separation between classes. Suspecting potential data leakage or an artificial pattern, we conducted a granular analysis of its distribution conditioned on the label.

Table III. URLSimilarityIndex Feature Statistics

Label	Count	Mean	Min	Max
Phishing (0)	60,31	49.65	0.16	67.29
Legitimate (1)	80,90	100.0	100.0	100.0

Table 3 reveals that `URLSimilarityIndex` exhibits zero variance for legitimate websites, with every observation fixed at 100.0. This creates a deterministic separation compared to the variable distribution of phishing URLs. While highly predictive, such a dominant feature risks masking the contribution of other predictors and leading to overfitting. Consequently, this feature will be excluded from the dataset to ensure a more robust and generalized model. The reasoning behind this is explained further in the data cleaning section.

Structural Data Assessment

1- Class Imbalance Check

An examination of the target variable (label) reveals a relatively well-balanced distribution. The train dataset consists of 85,214 legitimate URLs (57.3%) and 63,457 phishing URLs (42.7%).

With an imbalance ratio of 0.74, the minority class (phishing) is sufficiently represented. Consequently, this slight deviation does not constitute a severe imbalance, allowing us to proceed with modeling without the need for synthetic resampling techniques (e.g., SMOTE) or aggressive undersampling.

2- Assessment of Linearity (Box-Tidwell Approach)

Methodology: To evaluate the applicability of linear classifiers (e.g., Logistic Regression), we tested the linearity assumption between continuous features and the log-odds of the target variable. We utilized a Box-Tidwell-inspired approach, checking the correlation between predictors and their log-transformed interaction terms. We also attempted a log1p transformation to improve linearity.

Findings: The analysis revealed a predominantly non-linear feature space.

- Original State: Only 2 features satisfied the linearity condition.
- Post-Transformation: Applying a log transformation improved this count to 9 features (+7 improvement).

Interpretation of High Non-Linearity:

Despite the marginal improvement observed after log transformation, the vast majority of numerical features remain non-linear. This persistence indicates that the relationships between URL characteristics and phishing intent are complex and cannot be fully captured by simple linear equations. While we will proceed with Logistic Regression as a baseline model, this structural non-linearity suggests that the dataset is inherently better suited for non-linear algorithms (e.g., Random Forest or Gradient Boosting) to achieve maximum predictive performance.

3- Assessment of Normality (Shapiro-Wilk Test)

To verify distributional assumptions, we conducted the Shapiro-Wilk test on a random sample (N=300) of the numerical features. The analysis yielded p-values of approximately zero for all features except 2 features, leading to the rejection of the null hypothesis ($p < 0.05$). This confirms that the data follows a non-normal distribution.

Synthesizing visual and statistical findings, the dataset's inherent skewness reflects the adversarial nature of phishing, where URLs intentionally violate standard conventions. This structural non-normality acts as a distinct 'fingerprint,' guiding us to prioritize extreme deviations in the Feature Engineering phase.

4- Outlier Detection using IQR Method

Given the non-normal and right-skewed distribution of the numeric features, the Z-score method was deemed unsuitable. Instead, the Interquartile Range (IQR) method was selected as the most robust approach for detecting anomalies in this dataset. The analysis identified the highest outlier concentrations in `NoOfSubDomain` (24.55%), `Pay` (24.01%), and `IsHTTPS` (21.55%).

Instead of removing them, which would discard valuable predictive signals, the decision was made to keep these instances. This approach ensures the model remains sensitive to the extreme patterns often exhibited by phishing URLs.

5- Multicollinearity Analysis via VIF

The Variance Inflation Factor (VIF) analysis highlighted significant multicollinearity among the features, largely due to the structural nature of the derived attributes. Specifically, **6 features** exhibited severe multicollinearity with VIF values exceeding 10, while **7 features** showed moderate multicollinearity with VIF values between 5 and 10. Rather than discarding these high-VIF features and potentially losing critical information related to phishing patterns, we opted to retain them.

Table IV. Top 8 Feature VIF Scores

Feature Name	VIF Score
URLLength	1512.61
NoOfLettersInURL	785.97
NoOfDegitsInURL	79.26
NoOfOtherSpecialCharsInURL	36.42
DomainTitleMatchScore	17.52
URLTitleMatchScore	16.67
URLSimilarityIndex	9.39
NoOfObfuscatedChar	9.30

Table 4: *List of features exhibiting high Variance Inflation Factor (VIF) scores. Features with VIF > 10 indicate severe multicollinearity, while those between 5 and 10 indicate moderate multicollinearity. Due to their structural importance, these features are retained for dimensionality reduction via PCA.*

Research Questions

To guide the exploratory data analysis and subsequent modeling phases, this study formulates three primary research questions aimed at uncovering distinct behavioral and structural patterns in phishing URLs.

Question 1: Do phishing URLs exhibit a significantly higher frequency of numeric characters compared to legitimate websites?

The analysis shows a clear difference in the use of digits. Legitimate URLs contain almost no digits (mean ~ 0.05), while phishing URLs have an average of 4.27. This large gap suggests that a high digit count is a strong sign of phishing, likely used to hide the real domain.

Next, we examined the probability of observing specific digits (0-9) within the URLs.

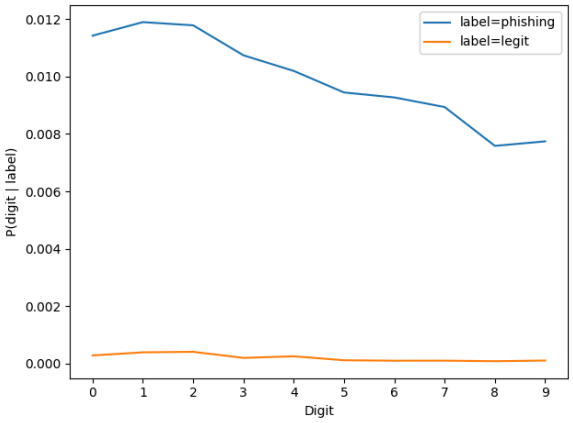

Fig 4. Probability of Phishing and Legitimate in URL

Conclusion: Figure 4 illustrates a clear distinction. For legitimate URLs, the probability of finding any digit is near zero (flat line). Conversely, phishing URLs exhibit a consistently high probability for all digits. This confirms that the presence of numbers—regardless of which digit it is—is a distinct characteristic of phishing attacks.

Question 2: Is there a statistically significant association between the presence of the 'www' prefix in a domain name and the website's classification as Phishing or Legitimate?

Hypothesis: Standard internet conventions typically include the 'www' subdomain for legitimate web services. We hypothesize that phishing actors may omit this prefix to shorten the URL or due to a lack of proper domain configuration.

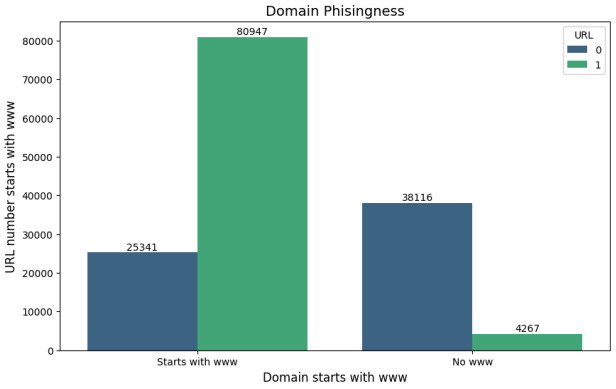

Fig 4. URL counts start with 'www'

1. Visual Analysis

The chart in Figure 4 shows a clear difference: legitimate websites almost always use the 'www' prefix (80,947 sites), while the 'No www' group is dominated by phishing URLs. Since very few legitimate sites lack this prefix (only 4,267), missing 'www' is a strong indicator of phishing.

2. Statistical Verification (Chi-Square Test)

To confirm this relationship statistically, a Chi-Square Test of Independence was performed on the contingency table.

Table V. Contingency Table

label	0	1
No 'www'	38116	4267
Starts with 'ww'	25341	80947

Conclusion: The test yielded a p-value of **0.000** ($p < 0.05$). Consequently, we reject the null hypothesis, confirming that a statistically significant dependency exists between the presence of 'www' and the class label. This validates the 'www' prefix as a critical binary feature for classification

Question 3: Does the variable '`TLDLegitimateprob`' represent the Target Encoding of the categorical '`TLD`' feature based on the legitimacy class?

Analysis & Conclusion: To empirically determine the encoding method, we analyzed the dataset's 605 distinct Top-Level Domains (TLDs). The investigation focused on identifying 'collisions' instances where different TLDs appear with the same frequency. The analysis revealed 45 such collisions, where TLDs shared the same frequency count yet were assigned different probability scores. This finding mathematically confirms that the feature does not rely on simple Count Encoding. Instead, the variable employs Target-Based Encoding, ensuring that the probability score reflects the specific legitimacy rate of each TLD rather than its raw frequency.

Table VI. Top 4 TLDs by Frequency and Probability

TLD	Frequency(Count)	Legitimate Probability
com	67,528	0.5229
org	11,290	0.0799
net	4,236	0.0384
uk	3,889	0.0285

V. Feature Engineering & Dimension Reduction

A. *Creating New Features*

We have created new features based on the domain of each URL.

- **`NoOfDigitsInDomain`**: Count of the digits in the domain name. Phishing domain addresses often include random characters such as digits.
- **`RatioOfDigitsInDomain`**: Ratio of digits to all characters in the domain name.
- **`NoOfHyphensInDomain`**: Count of the hyphen character to all characters in the domain name.
- **`HyphenRatioInDomain`**: Ratio of hyphens to all characters in the domain name.
- **`NoOfSpecialCharInDomain`**: Count of characters that are not alphanumeric in the domain name.
- **`AlphabetRatioInDomain`**: Ratio of alphanumeric characters to all characters in the domain name.
- **`NoOfTokensInDomain`**: Count of segments that are separated by a dot or a hyphen in the domain name.
- **`DomainEntropy`**: Shannon entropy of the domain name. It is a measure of unpredictability of the domain name [3].

The features `NoOfSelfRedirect` and `NoOfURLRedirect` only included binary values, although they are count variables. As such, we converted them to binary features named `HasSelfRedirect` and `HasURLRedirect`, respectively.

B. *Feature Transformation*

We applied power transformation methods in order to transform the numerical features. We picked a subset of the numerical features that are strictly positive and applied Box-Cox transformation. We applied Yeo-Johnson transformation to the remaining numerical features. To assess normality, we used skewness and kurtosis (Fisher's definition) statistics. Both metrics saw great reduction after the transformation. We also applied standard scaling at this step.

VI. Dimension Reduction

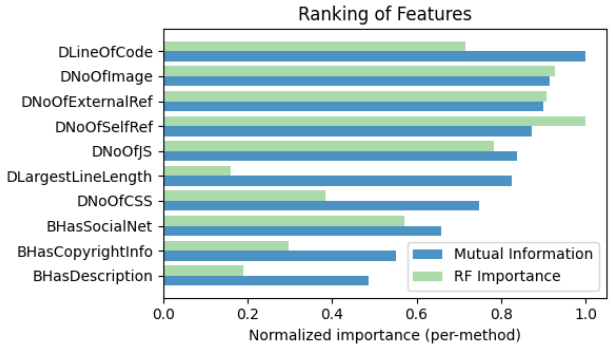

Fig 5. Top 10 ranked features by the two methods.

We applied several methods in order to understand the importance of each feature. Mutual information regression and random forest classification methods for feature selection both provide meaningful metrics that rank the features based on their importance. While some features had major differences in the ranking, both methods seem to agree on the most important features. In Figure 5, we have provided the rankings of the most important 10 features according to mutual information regression. Except for some features, the rankings are similar.

It is not surprising that features related to the complexity of a page rank higher. Phishing pages are often created very fast and without much effort. Also, no feature based solely on the URL ranks very high. The URL should also provide good information, but it may be the case that we were unable to extract the most important features from the URL. The highest ranked URL based feature by both methods was `TLD`. It is easy for attackers to obtain new and relatively unused TLDs, and the methods capture this phenomenon.

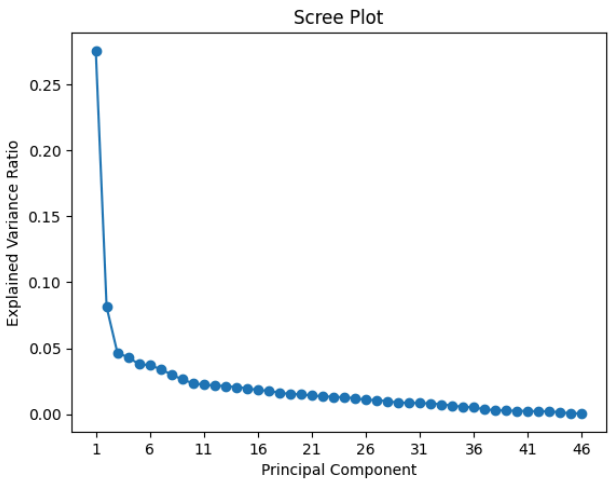

Fig 6. Scree plot to visualize explained variance ratio.

Next, we applied PCA on the transformed data. Before this step, we also applied transformation to the binary features. The explained variance ratio is visualized in Figure 6. We see that about 28% of the variation is explained by the first principal component. The first two components contribute a significant portion to the variance, and the remaining components do not contribute as much.

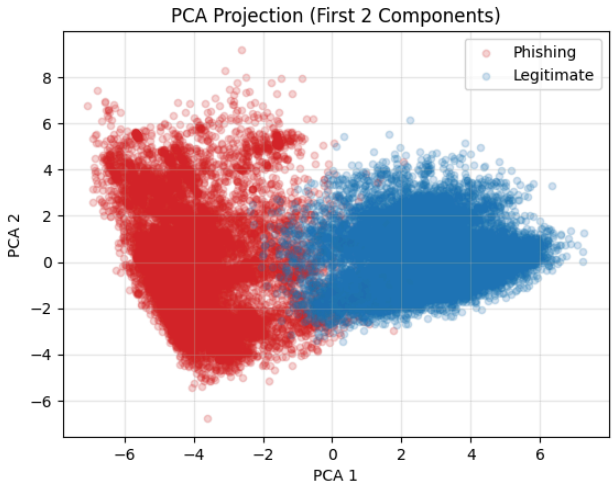

Fig 7. Visualization of the separation achieved by PCA.

In Figure 7, we have provided a scatter plot showing the separation achieved by the first two components of the PCA. We can see a separation between the two clusters, however, the border between the two clusters is fuzzy.

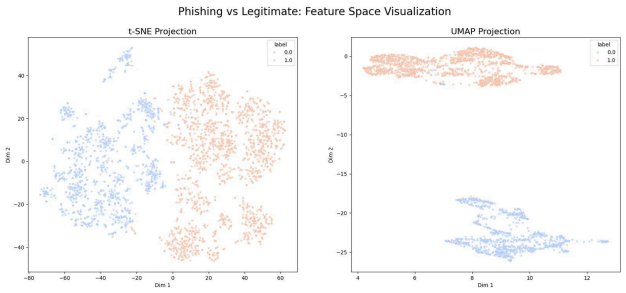

Fig 8. Visualization of clusters using t-SNE and UMAP

We also tried other methods such as t-SNE and UMAP, which are shown in Figure 8. But, due to limited computational power and high number of observations, we were unable to find optimal parameters for best visualizations. The ones shown use the default parameters provided by `scikit-learn` and `umap-learn` packages in Python. We see that UMAP captures local variation very well and both of the approaches are able to cluster the labels well. However, there are some errors in both cases, more prominent in t-SNE.

VII. Statistical Modeling & Conclusions

Setup:
We split our data as a train set 80% of the total data set and test set 20% for the total data set. We use a validation test for k-fold-cross-validation. We pick k as a 5 and part of the Logistic RegressionCV we split our train 5 equal parts and validated for 5 times. Each time we use one part to split 5 equal parts. Specifically we pick part 1,2,3,5 for training and use part 5 for validation. Consequently we end up with 5 validated results and use all train sets for validation as a result it gives average hyperparameters results for all validation results. A fixed random seed (*random_state42*) was used for all models to ensure the used validation tests were identical. Fixing the seed allows us a fair comparison between Base, Ridge, Lasso and ElasticNet Models.

Model Selection:

1. **The Nature of the Dependent Variable:**

Our data has binary features and to be able to use linear regression we need to have numeric features only we justify in binary features to 0(no) or 1(yes) . We cannot apply linear regression on these features. The reason we use logistic regression for binary outcomes. Logistic Regression gives us 2 categoric outcomes (binary format).

2. **Bounded Probabilities:**

The problem of Linear Regression predicts the value as -0.2, 2.5 , 0.6 in our case target feature is binary. Therefore giving predicted values on our target format is not acceptable.

- What does it mean to have a 250% chance of being a legitimate website? It is mathematically impossible.

Sigmoid(Logistic) Function transforms the outcome to binary.

- It squashes the result strictly 0 and 1 to give us a clear legitimate or phishing website class.

3. **Violation of Homoscedasticity:**

Linear Regression assumes constant variance (homoscedasticity). In our case we have binary features which violate the constant variance assumptions. Binary features means that variance depends on means so when means change error variance changes as probability changes. . The reason we do not use Linear Regression is that Least Squares cannot handle the variances, we use logistic Regression to handle it by Maximum Likelihood Estimation.

Assumption Check:

- **Linearity of the Logit:** The assumption of a linearity between continuous independent variables and log-odds of the dependent variable used in the Box-Tidwell test and visual inspection of scatter plots.
  - To satisfy this assumption we use `PowerTransformation` which was integrated into the pipeline. This transformation automatically applies correct transformation.

- **Multicollinearity:** High correlation among features was identified using VIF (Variance Inflation Factor) analysis in the EDA part. To handle Multicollinearity we apply two methods and compared:
  1. **Dimensionality Reduction:** We applied PCA(Principal Component Analysis) to transform correlated data to independent components.
  2. **Regularization:** Lasso(L1) and Ridge(L2) regressions were applied. The L1 method helps us to eliminate low effect features and the L2 method helps us to shrink the coefficient of the features and fix the multicollinearity problem. This is in line with recent research that has found that using Regularized Logistic Regression is effective at handling correlated features to preserve the model performance metrics ability[4].

- **Influential Outliers:** Potential outliers are detected by IQR method . observations falling outside the range of $Q1 - 1.5xIQR$ and $Q3 + 1.5xIQR$.
    - Since extreme values crucial in phishing features (e.g. extreme long URLLength) are often related to phishing websites. Therefore these values were **not removed** to avoid loss of critical fraud information. Instead, the Power Transformation (Yeo-Johnson) was applied in the pipeline. This method suppressed their dominance, preventing them from distorting the model coefficients while keeping the information they have.

- **Imbalance Data:** We check in the EDA part balance of the target feature to avoid imbalance data. We observe that the target features have 57% legitimate websites and 43% phishing websites. In the vision of this finding our data is balanced and we **do not** need to apply techniques such as **SMOTE**.

VIII. Model Evaluation and Conclusion

1. Performance Evaluation Metrics

To find out the effectiveness of the phishing detection models, multiple metrics examine the best model rather than the standard accuracy. The models were evaluated based on **Recall(Sensitivity) Specificity, F1-Score, Cohen's Kappa and ROC AUC.**

Table VII. PERFORMANCE METRICS OF LOGISTIC REGRESSION PIPELINES

Model Strategy	Performance Metrics					
	Accuracy	F1-Score	Sensitivity	Specificity	Kappa	ROC-AUC
Base Logistic	0.997322	0.997659	0.997477	0.997114	0.994530	0.999956
Lasso (L1)	0.997351	0.997684	0.997477	0.997184	0.994591	0.999958
Ridge (L2)	0.997307	0.997646	0.997529	0.997010	0.994499	0.999955
Elastic Net	**0.997381**	**0.997711**	**0.997555**	**0.997149**	**0.994651**	**0.999957**
PCA + Base	0.995596	0.996149	0.995864	0.995236	0.991005	0.999861
PCA + L1	0.995536	0.996097	0.995786	0.995202	0.990884	0.999863

The performance comparison of the **Base Logistic Model**, **Regularized Models (Lasso, Ridge, Elastic Net)**, and **PCA-Transformed Models** is presented in **Table 7**. Performances of the models are exceptionally high which means that all our models perform well. In this comparison best overall performers are **ElasticNet** with an F1 score 0.9977 and an ROC-AUC of 0.9999.

- **Sensitivity (0.9976):** The model's ability to identify correctly to Legitimate websites. The high value of the sensitivity ensures that valid websites are not mislocated (Low False Positive Rate).
- **Specificity (0.9971):** This is the primary security objective to detect phishing websites by our model (True Negative Rate).

2. Comparison of Modeling Strategies

- **Regularization Effects:** The best model ElasticNet achieves that 0.9977 F1 score. Comparison with the Base Logistic model, regularization avoids the overfitting by penalizing large coefficients. Specifically, Lasso model reduced the number of the feature by sharing exactly zero, acting like an embedded feature selection method.Meanwhile Ridge eliminated the low coefficient features and generated a more compact model. ElasticNet combines the Lasso and Ridge approach and outperforms every model.
- **Dimensionality Reduction (PCA):** The model trained on PCA (Principal Component Analysis) components achieve that 0.9960 F1 score. While PCA effectively resolved multicollinearity by generating orthogonal components, it resulted in a slight negative in performance compared to the other model such as Lasso. Additionally, the PCA model's original web features were not directly interpretable which makes the model a "black box" and not able to see meaningful distinction of the original features even you are able to see the each PCA coefficient by original features but it make it too complex to identify the PCA meanings.This corroborates the finding of a study conducted by Alnemari and Alshammari, stating that feature selection techniques can comparability perform PCA in the process of phishing identification in terms of important threat indicators for performance metrics[5].
- **Imbalance Handling:**By applying class_weight='balanced' (it makes it cost sensitive to model), the models maintain high Sensitivity (average 0.9969) , ensuring that phishing websites are correctly identified, which is the case of this paper.

3. Coefficient Interpretation & Logical Validation

TABLE VIII. TOP INFLUENTIAL FEATURES (ELASTIC NET)

Feature	Coefficient	Abs_Coef
IsHTTPS	5.497374	5.497374
LineOfCode	3.383056	3.383056
HasDomainTitleMatch	2.792139	2.792139
AlphabetRatioInDomain	-2.482183	2.482183

HasDigitInDomain	2.358955	2.358955
HasSocialNet	2.033317	2.033317

To understand the model's logic, the coefficients of the best performance model were analyzed. As shown in Table 8, the direction of the relationships corresponds to domain expertise:

- **Positive Associations:** Features such as `IsHTTPS` and `LineOfCode` had significant positive coefficients. `IsHTTPS` has the most influential effect on legitimacy. It is consistent with domain information, as niche websites almost always employ a valid SSL/TLS encryption method[6]. `LineOfCode` had the second most influential feature for being a legitimate website. This shows that the complexity of the website is important for classing the legitimate website. also it indicates that phishing websites had less lines of code which means that phishing websites are not complex as much as legitimate websites.
- **Negative Associations:** In our model most of the features support legitimate websites by positive effect of coefficient. However, there are still highly influenced negative effects. `AlphabetRatioInDomain` (-2.48) indicates that higher the alphabet ratio is, lesser the probability of a phishing attack is.
- **Validation Conclusion:** These coefficients logical consistency shows that the model is accurately capturing the features of the phishing websites rather than learning noise of the features. It means that our model is valid for real life phishing websites and will perform well.

IX. Conclusion

In this study, we find a promising result to classify phishing and legitimate websites. Our overall model performs really well compared to general benchmarks. This 0.99 finding is not new for the phishing model. However, we developed a comparable model using logistic regression with the other researchers[7]. Handling multicollinearity and non-normality in data, this research was able to establish an effective classification model. Also, the assumption for linearity was checked for four features (`TLD`, `URLCharProb`, `AlphabetRatioInDomain`, `DomainEntropy`) finding that one of the features `URLCharProb` does not satisfy linearity. However we want to observe models' performances with that feature included. We found that it was not among the top 10 most influential coefficients. In the light of that result we decided to keep the feature. Through Variance Inflation Factor (VIF) analysis and Power Transformations, we created a highly effective classification model.

Our experimental results show that ElasticNet Logistic Regression achieved the best performance overall, where the results for performance metrics were 0.9977 for F1-Score and 0.9999 for ROC-AUC. This almost perfect capacity extract model can effectively distinguish between legitimate and phishing domains without complex "black box" algorithms.

Key findings from this research include:

Regularization vs. Dimensionality Reduction: Analysis of our findings indicates that regularization techniques outperformed PCA across all performance metrics. The reason could be that we selected a threshold of 85% explained variance which means that we lost some meaningful nuance from the dataset. However PCA successfully reduced computational power needs, which should be taken into consideration. PCA successfully reduced dimension 44 to 21 while remaining 85% of the information explained.

Structural and Behavioral Indicators: Our model confirms that domain knowledge highly influences the prediction of phishing websites. Positive coefficient indicates that legitimate websites characteristics have patterns in `IsHTTPS` and `LineOfCode`. This is explained by still having a https license that they have significant character of legitimate websites. Also, the line of code in the website explaining to us more code in the website has a better chance to be legitimate. In that finding, increasing complexity of the websites is a great tool to detect legitimate websites, which means that phishing websites have less complex, hastily written, less lines of code.

Model Robustness: By applying class weight balance to models we ensure that sensitivity (0.9976) effectively minimizes the risk of false negatives, a most critical metric for cybersecurity systems where missing the phishing websites can have several consequences.

In conclusion, this study demonstrates that a regularized linear approach yields effective results with respect to preserving model explainability "white box" and accuracy. It successfully captures and understands the nature of the phishing URLs while proving the logical real world security applications. Future improvement may explore the rest of the misclassified phishing and legitimate classes and search patterns for these false negative and positive values. Future work may involve feature creation to capture the rest of the false classified values to outperform existing ElasticNet Logistic Regression.

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